

Q u e s t i o n s

1. What section of the online user's manual and command reference holds administrative (not user) commands? Select one.

1. Section 1
2. Section 2
3. Section 8
4. Section n
5. Section s

2. In the bash shell, entering the !! command has the same effect as which one of the following?

1. Ctrl-P and Enter
2. Ctrl-N and Enter
3. Ctrl-U and Enter
4. !-2
5. !2

3. Name the command that displays pages from the online user's manual and command reference.

4. Which of the following commands displays the comments from a bash script? Select all that apply.

1. find "^#" /etc/rc.d/rc.local
2. sed '/^#/ !d' /etc/rc.d/init.d/httpd
3. grep ^# /etc/rc.d/init.d/httpd
4. grep ^# /etc/passwd
5. locate "^#" /etc/skel/.bashrc

5. Which one of the following answers creates an environment variable VAR1, present in the environment of a bash child process?

1. VAR1="fail" ; export VAR1
2. VAR1="fail" \ export VAR1
3. VAR1="fail"
4. set VAR1="fail" ; enable VAR1
5. export VAR1 \ VAR1="fail"

6. Name the full path and name of the file that holds most of the information on system user groups.

7. Which one of the following outcomes results from the following command?

```
# chmod g+s /home/software
```

1. The SUID bit will be set for /home/software.
2. The SGID bit will be set for /home/software, preventing access by those not a member of the software group.
3. The SGID bit will be set for /home/software, to keep group membership of the directory consistent for all files created.
4. The sticky bit will be set for /home/software.
5. The sticky bit will be applied to all files in /home/software.

8. Consider the following script, stored in a file with proper modes for execution:

```
#!/bin/bash
for $v1 in a1 a2
do
echo $v1
done
```

Which one of the following best represents the output produced on a terminal by this script?

1. in a1 a2
2. ala2
3. \$v1 \$v1 \$v1
4. No output is produced, but the script executes correctly.
5. No output is produced, because the script has an error.

9. Which one of the following commands verbosely extracts files from a tar archive on a magnetic tape device?

1. tar cvf /dev/st0
2. tar cvf /dev/ttyS0
3. tar xvf /dev/st0
4. tar xvf /dev/ttyS0
5. tar rvf /dev/st0

10. Alex wants to protect himself from inadvertently overwriting files when copying them, so he wants to alias cp to prevent overwrite. How should he go about this? Select one.

1. Put alias cp='cp -i' in ~/.bashrc.
2. Put alias cp='cp -i' in ~/.bash_profile.
3. Put alias cp='cp -p' in ~/.bashrc.
4. Put alias cp='cp -p' in ~/.bash_profile.
5. Put alias cp = 'cp -I' in ~/.bashrc.

11. Monica consults the /etc/passwd file expecting to find encrypted passwords for all of the users on her system. She sees the following:

```
jdoe:x:500:500:~/home/jdoe:/bin/bash
bsmith:x:501:501:~/home/bsmith:/bin/tcsh
```

Which of the following is true? Select one.

1. Accounts jdoe and bsmith have no passwords.
2. Accounts jdoe and bsmith are disabled.
3. The passwords are in /etc/passwd-.
4. The passwords are in /etc/shadow.
5. The passwords are in /etc/shadow-.

12. What variable holds the list of directories searched by the shell to find executable programs?

13. Where does The Linux System Administrators' Guide originate? Select one.

1. Red Hat Software, Inc.
2. O'Reilly Media, Inc.
3. The Free Software Foundation
4. The Linux Documentation Project
5. Usenet newsgroup comp.os.linux

14. Which one of these files determines how messages are stored using syslogd?

1. /etc/sysconfig/logger.conf
2. /etc/syslog.conf
3. /etc/syslogd.conf
4. /etc/conf.syslog
5. /etc/conf.syslogd

15. How many hosts can exist on a subnet with mask 255.255.255.128? Select one.

1. 512
2. 256
3. 128
4. 127
5. 126

16. When running a text-mode FTP client, which command retrieves multiple files? Select one.

1. get *.txt
2. retrieve *.txt
3. mget *.txt
4. mretrieve *.txt
5. get -m *.txt

17. For an Internet workstation with a single network interface, what routes must be added to interface eth0 after it is initialized? Select one.

1. None
2. Interface
3. Interface and default gateway
4. Interface, local network, and default gateway

18. On a Linux server, what service is most likely "listening" on port 25? Select one.

1. Apache
2. News
3. Sendmail
4. Samba
5. FTP

19. Which one of these protocols is used as a datagram delivery service by the remaining three? Select one.

1. TCP
2. IP
3. UDP
4. ICMP

20. Which command will display information about Ethernet interface eth0? Select one.

1. cat /proc/eth/0
2. ifconfig eth0
3. ipconfig eth0
4. ipconfig /dev/eth0
5. cat /etc/eth0.conf

21. When is the PPP interface ppp0 created? Select one.

1. At boot time by the kernel
2. At installation time by mkndod
3. At dial-up time by the chatscript
4. At dial-up time by pppd
5. When the modem powers up

22. What program is run on a client machine to request an IP address from a DHCP server? Select one.

1. dhcpd
2. inetd
3. pump
4. dhcp_client
5. bootp

23. What does the printcap entry sd indicate? Select one.

1. The system default printer
2. A printer's spool directory
3. A device file for the printer
4. A location where errors are stored
5. The printer driver

24. Where is TCP wrappers configured and where is it enabled?

1. Configured in tcpd.conf, enabled in tcpd.conf
2. Configured in inetd.conf, enabled in inetd.conf
3. Configured in hosts.deny and hosts.allow, enabled in inetd.conf
4. Configured in inetd.conf, enabled in hosts.deny and hosts.allow

25. Which of the following is a valid entry in /etc/fstab for a remote NFS mount from server fs1? Select one.

1. fs1:/proc /mnt/fs1 nfs defaults 9 9
2. /mnt/fs1 fs1:/proc nfs defaults 0 0
3. fs1:/home /mnt/fs1 nfs defaults 0 0
4. /mnt/fs1 fs1:/home nfs defaults 0 0
5. /home:fs1 /mnt/fs1 nfs defaults 0 0

26. Which network protocol is used by Telnet and FTP? Select one.

1. ICMP
2. UDP
3. TCP
4. DHCP
5. PPP

27. Which of the following programs will display DNS information for a host? Choose all that apply.

1. host
2. nslookup
3. nsstat
4. dig
5. ping

28. Consider the following entry in /etc/exports:

```
/home          pickle(rw,no_root_squash)
```

How is this entry handled by the NFS daemon? Select one.

1. Directory /home is shared to everyone, without requiring passwords.
2. Directory /home is shared to everyone, requiring passwords.
3. Directory pickle is mounted on /home.
4. Root is not allowed access to the shared directory.
5. The mount attempt will fail.

29. From the user's point of view, which answer describes the appearance of an NFS mounted directory? Select one.

1. A new device in /dev.
2. A new local volume accessed using a volume letter, such as D:.
3. A new local volume accessed using the NFS server's name.
4. Part of the local filesystem, accessed using ordinary pathnames.
5. Part of the NFS server's filesystem, accessed using the NFS server's name.

30. Which of the following statements regarding the ICMP protocol is not true? Select one.

1. ICMP is connectionless.
2. ICMP provides network flow control.
3. ICMP is also known as UDP.
4. ICMP is used by ping.

31. What is CHAP? Select one.

1. The PPP chat script.
2. An authentication protocol using clear text.
3. An authentication protocol embedded in the PPP data stream.
4. The pppd configuration utility.
5. A modem communications protocol.

32. What server daemon resolves domain names to IP addresses for requesting hosts?

33. During the two-way communication that takes place during a chat script used to start PPP, what is chat communicating with? Select one.

1. The pppd daemon
2. The PPP server
3. The kernel
4. The modem
5. The syslogd daemon

34.What function does a print filter serve? Select one.

1. It collates output from multiple users.
2. It translates various data formats into a page description language.
3. It rejects print requests from unauthorized users.
4. It rejects print requests from unauthorized hosts.
5. It analyzes print data and directs print requests to the appropriate lpd.

35.Consider the following excerpt from file /etc/resolv.conf of a Linux workstation:

```
nameserver 127.0.0.1
nameserver 192.168.1.5
nameserver 192.168.250.2
```

What can be said about this configuration? Select one.

1. Two DNS servers on the public network are being used for resolution.
2. One DNS server on the local network is being used for resolution.
3. The configuration contains errors that will prevent the resolver from functioning.
4. A caching-only name server is running.
5. The resolver library will consult nameserver 192.168.250.2 first.

36.Which of the following is true regarding BIND v4 and BIND v8 configuration files? Select one.

1. The information is largely the same, but the syntax is different.
2. The syntax is largely the same, but the information is different.
3. The two BIND versions use the same configuration file.
4. BIND v4 uses a binary configuration file instead of text.
5. BIND v8 uses a binary configuration file instead of text.

37.What file is used to configure Sendmail? Include the entire path.

38.Name the file that contains simple mappings between IP addresses and system names.

39.What is the meaning and location of the following kernel configuration file excerpt? Select one.

```
options opl3 io=0x388
```

1. Kernel option opl3 is set to use I/O port 0x388; /usr/src/linux/.config.
2. Kernel module option opl3 is set to use I/O port 0x388; /usr/src/linux/.config.
3. Kernel module opl3 is set to use I/O port 0x388; /usr/src/linux/.config.
4. Kernel option opl3 is set to use I/O port 0x388; /usr/src/linux/.config.
5. Kernel module opl3 is set to use I/O port 0x388; /etc/conf.modules or /etc/modules.conf.

40. What program can be used to interactively change the behavior of a print queue? Select one.

1. lpd
2. lpr
3. lpq
4. lprm
5. lpc

41. Which of the following represents a valid sequence of commands to compile and install a new kernel? Select one.

1. make modules_install; make modules; make bzImage; make clean; make dep
2. make dep; make clean; make bzImage; make config; make modules; make modules_install
3. make config; make dep; make clean; make bzImage; make modules; make modules_install
4. make config; make bzImage; make dep; make clean; make modules; make modules_install
5. make dep; make clean; make bzImage; make modules; make modules_install; make config

42. What program will display a list of each hop across the network to a specified destination? Select one.

1. tracet
2. rttrace
3. traceroute
4. routetrace
5. init

43. Which file holds configuration information used during the process of kernel compilation? Select one.

1. /usr/src/linux/config
2. /usr/src/linux/.config
3. /usr/src/linux/kernel.conf
4. /etc/kernel.conf
5. /etc/sysconfig/kernel.conf

44. After a PPP connection is established and authenticated, what needs to be done before the interface can be used? Select one.

1. Add a route to ppp0.
2. Enable ppp0.
3. ifup ppp0.
4. Run pppd.
5. Turn on the modem.

45. Which of the following is not the name of an Apache configuration file? Select one.

1. httpd.conf
2. html.conf
3. srm.conf
4. access.conf

46.Which statement is true regarding the configuration of a printer on a remote Windows machine? Select one.

1. It can be configured like a TCP/IP network-attached printer.
2. The input filter must be set to smbprint.
3. The Windows printer must contain PostScript capability.
4. The rp directive must be used in the printcap file.
5. Linux can't print to Windows printers.

47.What types of files are located in the directory tree specified by the Apache DocumentRoot configuration directive? Select one.

1. Apache documentation files.
2. Apache configuration files.
3. Web site HTML files.
4. Web site configuration files.
5. Apache startup and shutdown commands.

48.Which of the following commands will display a listing of files contained in a tar archive tape in /dev/st0? Select one.

1. tar cf /dev/st0
2. tar xf /dev/st0
3. tar tf /dev/st0
4. tar -zf /dev/st0
5. tar -zcvf /dev/st0

49.What is the systemwide bash configuration file called? Include the entire path.

50.How can a nonprivileged user configure sendmail to forward mail to another account? Select one.

1. She can add a new entry in /etc/aliases.
2. She can create a .forward file containing the new address.
3. She can create an .alias file containing the new address.
4. She can create a sendmail.cf file containing the new address.
5. She cannot forward mail without assistance from the administrator.

51.How does a process indicate to the controlling shell that it has exited with an error condition? Select one.

1. It prints an error message to stderr.
2. It prints an error message to stdout.
3. It sets an exit code with a zero value.
4. It sets an exit code with a nonzero value.
5. It causes a segmentation fault.

52.Consider the following trivial script called myscript:

```
#!/bin/bash
echo "Hello"
echo $myvar
```

Also consider this command sequence and result:

```
# set myvar='World'
# ./myscript
Hello
```

The script ran without error but didn't echo World. Why not? Select one.

1. The syntax of the set command is incorrect.
2. The script executes in a new shell, and myvar wasn't exported.
3. The `#!/bin/bash` syntax is incorrect.
4. The `$myvar` syntax is incorrect.
5. The script is sourced by the current shell, and myvar is available only to new shells.

53. What does this short configuration file excerpt tell the Samba daemon? Select one.

```
[home]
    path = /home
    guest ok = yes
    writable = yes
```

1. The location of the Samba software is rooted at /home.
2. A printer called home uses /home as a spool directory.
3. A share called home is located on /home and is writable by authenticated users.
4. A share called home is located on /home and is writable by anyone.
5. A share called home on remote system guest will be mounted at /home.

54. How can the finger daemon be enabled? Select one.

1. Uncomment the `in.fingerd` line in `/etc/inetd.conf`.
2. Use cron to run fingerd once per minute.
3. Include fingerd in the TCP wrappers configuration.
4. Remove fingerd from `hosts.deny`.
5. Add fingerd to `hosts.allow`.